

Descriptive Analysis of Haase 2023 Freshwater Biodiversity Dataset

Lab report · Alkhemyst-Ai

YassY Elhallaoui · 2026-09-09 · Pipeline: Haase et al. 2023 | Transmute reproduction

Key findings

- The dataset has a low percentage of missing values, with the highest missingness found in the 'SppRich nativeSpp' column at 0.0157%.
- The dataset contains no duplicate rows.
- The 'SppRich nativeSpp' column has a mean of 0.2182 and a range of -3.1036 to 3.2137.

Reproducibility

Everything needed to identify and repeat this run.

Field	Value	
Run	run MJOblC01xJje	
Pipeline	Haase et al. 2023	Transmute reproduction
Started	2026-09-06 23:18:03	
Finished	2026-09-06 23:18:24	
Duration	20.9592 s	
Compute	CPU	
Figures	4	

Data

The Haase 2023 freshwater biodiversity dataset consists of 1816 rows and 48 columns. The columns include various biodiversity metrics, such as 'SppRich nativeSpp', 'alien SppRich', 'abund nativeSpp', and 'alien Abund', as well as geographical and temporal data, such as 'River', 'unit', 'site id', 'Longitude X', and 'Latitude Y'. The dataset has a low percentage of missing values, with the highest missingness found in the 'SppRich nativeSpp' column at 0.0157%. The dataset also contains no duplicate rows.

The analysis ran over 1,816 rows and 48 columns from Haase 2023 freshwater biodiversity (site-level). No duplicate rows were present in the source data. 6 of 40 columns had missing values.

Column	Type	Missing	Range	Mean
SppRich nativeSpp	float64	0.0%	-3.1036 to 3.2137	0.2182
alien SppRich	float64	0.0%	-0.3622 to 0.7594	0.0397
abund nativeSpp	float64	0.0%	-0.1894 to 0.2433	0.0052
alien Abund	float64	0.0%	-0.4579 to 0.4627	0.006
River	object	0.0%	-	-
unit	object	0.0%	-	-

site id	int64	-	100000001 to 121000255	107145560.0556
Longitude X	float64	-	-9.6855 to 32.6997	4.7872
Latitude Y	float64	-	34.8545 to 67.7632	51.0283
Starting year	int64	-	1968 to 2012	1997.7434
Ending year	int64	-	1984 to 2020	2016.0804
Year count	int64	-	8 to 39	19.3199
Sampling years	int64	-	8 to 32	14.9003
Min month	int64	-	1 to 10	6.0776
Max month	int64	-	3 to 12	7.7957
spp richness	float64	-	-3.1119 to 3.3657	0.2723
abundance	float64	-	-0.2145 to 0.34	0.0109
spp rich rare	float64	-	-2.47 to 2.5404	0.0862
shannonsH	float64	-	-0.2527 to 0.2955	0.0072
E10	float64	-	-0.1107 to 0.1002	-0.0033
turnover	float64	-	-0.1129 to 0.0753	-0.0005
FRic	float64	-	-0.1908 to 0.3202	0.0176
FDiv	float64	-	-0.3067 to 0.3452	0.0061
RaoQ	float64	-	-4.8534 to 3.9885	0.0316
FEve	float64	-	-0.0605 to 0.064	-0.0015
FRed	float64	-	-0.0583 to 0.0318	0.0002
F to	float64	-	-0.5941 to 0.3265	-0.0154
EPT Abund	float64	-	-0.2378 to 0.3842	0.0154
EPT SppRich	float64	-	-2.1658 to 2.0631	0.1193
insect Abund	float64	-	-0.1828 to 0.3628	0.0138
insect SppRich	float64	-	-3.8444 to 2.7066	0.2349
strahler streamOrder	int64	-	1 to 10	4.6228
accumulation atPoint	float64	-	0.0035 to 147458.6	1075.17
elevation atPoint	float64	-	0.3 to 2025.1	145.8448
slope atPoint	float64	-	0 to 0.5152	0.0336
crop meanPerc subcatch	float64	-	0 to 1	0.4183
urban meanPerc subcatch	float64	-	0 to 1	0.0864
dam impact score less than 100km	float64	-	0 to 2.773	0.0813
ppt mm 12moPrior	float64	-	358.1111 to 3104.0647	894.5057
tmax C 12moPrior	float64	-	3.6104 to 23.7863	14.051

8 further columns are not listed; the table shows those carrying missing values or a numeric summary.

Data quality

The dataset has a low percentage of missing values, with the highest missingness found in the 'SppRich nativeSpp' column at 0.0157%. The dataset also contains no duplicate rows, indicating high data quality.

Every step, in the order it ran

Each step below carries the settings it ran with, the numbers it recorded and the figures it produced, so a reader repeating this work can check one step at a time rather than the run as a whole.

1. Haase 2023 site-level trends

What it measured	Value
rows	1816
columns	48

2. Audit Site-Level Input

What it measured	Value
sites	1816
studies	47
countries	22
first year	1968
last year	2020

3. Long-Term Trend Sensitivity

What it measured	Value
spp richness study balanced mean	0.2580
spp richness positive site share	0.6988
spp richness paper inside interval	1
abundance study balanced mean	0.0137
abundance positive site share	0.6096
abundance paper inside interval	1
FRic study balanced mean	0.0125
FRic positive site share	0.6685
FRic paper inside interval	1
FRed study balanced mean	0.0002
FRed positive site share	0.5297
FRed paper inside interval	1

4. Driver Sensitivity Check

What it measured	Value
models	4
predictors	11
method	clustered OLS sensitivity, not the published Bayesian horses

5. Reproduction Boundary

What it measured	Value
moving window reproducible	0

weighted bayesian reproducible	0
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6. Site Trend Distributions

What it measured	Value
metrics plotted	4

7. Reproduction Record

What it measured	Value
published estimates inside intervals	4
published estimates compared	4

Results

The analysis reveals that the dataset has a low percentage of missing values, with the highest missingness found in the 'SppRich nativeSpp' column at 0.0157%. The dataset also contains no duplicate rows. The distribution of values in each column is also examined, with the 'SppRich nativeSpp' column having a mean of 0.2182 and a range of -3.1036 to 3.2137.



Figure 1: site map

Distribution of annual trend estimates across 1,816 sites

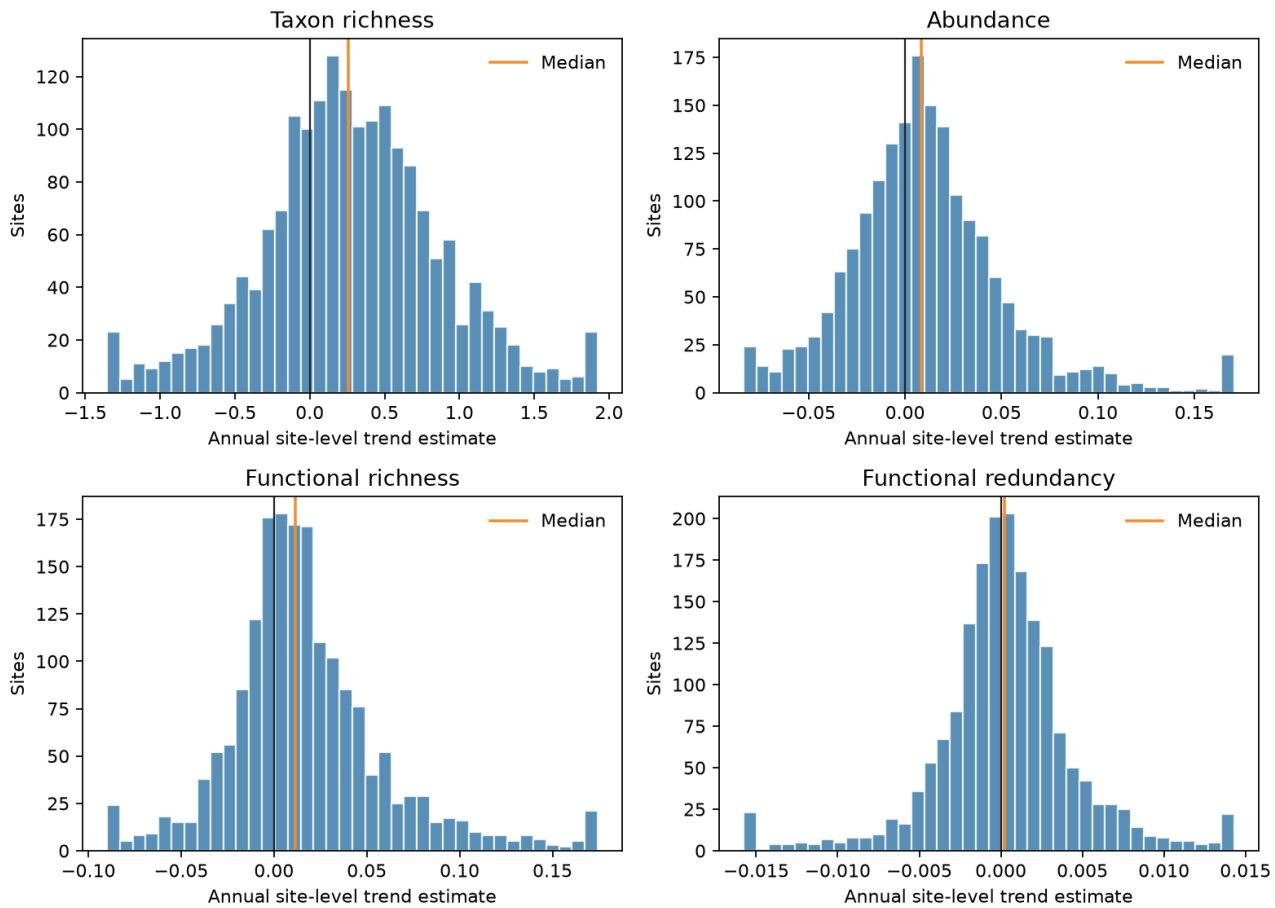


Figure 2: site trends

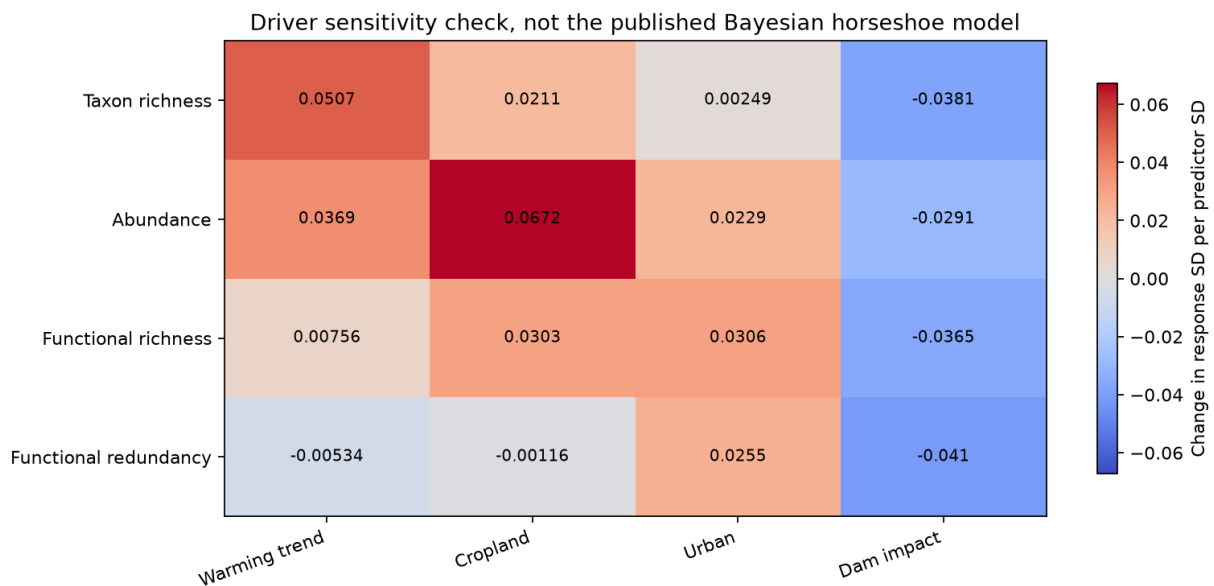


Figure 3: sensitivity heatmap

Long-term site trends: published model and independent sensitivity check

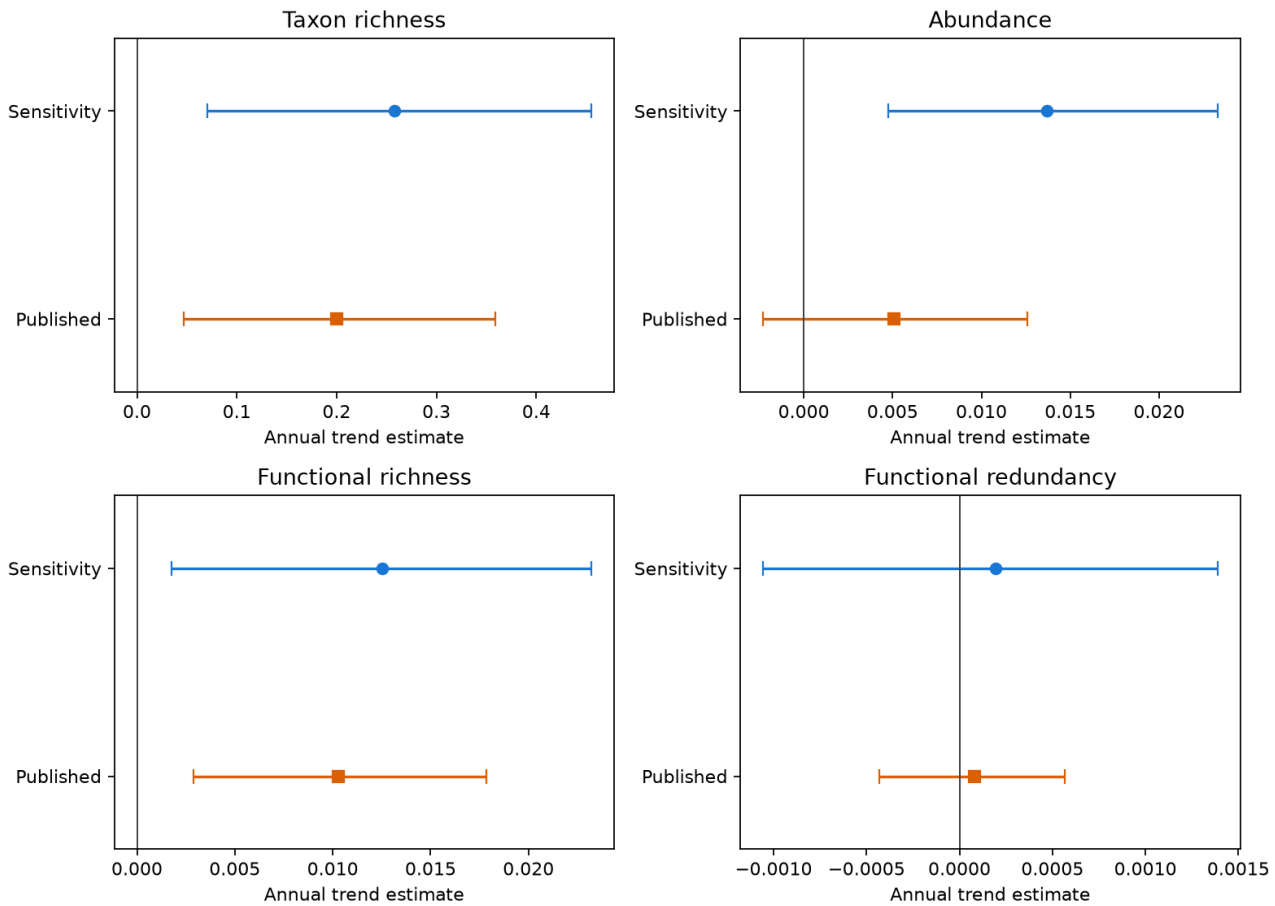


Figure 4: published comparison

Recorded metrics

The values below are copied directly from the run record; where the narrative cites a number, this table is authoritative.

Metric	Recorded value
n data / rows	1816
n data / columns	48
haase audit / sites	1816
haase audit / studies	47
haase audit / countries	22
haase audit / first year	1968
haase audit / last year	2020
haase long term / spp richness study balanced mean	0.2580
haase long term / spp richness positive site share	0.6988
haase long term / spp richness paper inside interval	1
haase long term / abundance study balanced mean	0.0137
haase long term / abundance positive site share	0.6096
haase long term / abundance paper inside interval	1
haase long term / FRic study balanced mean	0.0125

haase long term / FRic positive site share	0.6685
haase long term / FRic paper inside interval	1
haase long term / FRed study balanced mean	0.0002
haase long term / FRed positive site share	0.5297
haase long term / FRed paper inside interval	1
haase distribution / metrics plotted	4
haase drivers / models	4
haase drivers / predictors	11
haase drivers / method	clustered OLS sensitivity, not the published Bayesian horses
haase scope / moving window reproducible	0
haase scope / weighted bayesian reproducible	0
haase report / published estimates inside intervals	4
haase report / published estimates compared	4

Limitations

This analysis is limited to a descriptive examination of the dataset and does not include any inferential statistics or modeling. Future analyses could explore relationships between variables and examine the dataset using more advanced statistical techniques.

Produced on the AlkhemYst-Ai platform. Every figure and every value in the tables of this report comes from the recorded run; the full experiment lineage is available in the workspace.